Executive Summary: Cell-Type-Specific Biomarkers in Human Liver Fibrosis

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Objective

Discover and prioritize cell-type-specific biomarkers and therapeutic targets in human liver fibrosis from single-cell transcriptomics (primary: GSE136103, cirrhotic vs healthy; validation: GSE244832, MASLD/MASH F0 to F4), focusing on the hepatic stellate/myofibroblast, macrophage/monocyte, and endothelial compartments.

Approach

A reproducible Snakemake + Scanpy pipeline ingests and QCs the data, integrates over donor (not disease) and verifies the fibrosis signal is preserved, annotates and validates the three disease compartments, runs differential expression, characterizes mechanism (pathway/TF activity and ligand-receptor analysis with downstream-target corroboration), and prioritizes candidates with a transparent composite score (DE effect, cell-type specificity, cross-dataset reproducibility, druggability, biomarker accessibility, and an explainable XGBoost/SHAP signal). Two choices keep the ranked list trustworthy rather than a marker re-discovery: differential expression is donor-aware pseudobulk (the replication unit is the donor, not the cell, which avoids the pseudoreplication that inflates cell-level tests), and the composite is gated so a gene must be statistically DE-significant (at the compartment or disease-subpopulation level) and cell-type-specific before specificity, druggability, or accessibility can lift it; abundant lineage markers and ambient cross-compartment leakage are filtered out.

Top candidates

	rank	gene	compartment	composite	accessibility_class	druggability
0	1.0	PDGFRA	stellate_myofibroblast	0.671	surface	0.90
1	2.0	ACKR1	endothelial	0.635	surface	0.30
2	3.0	LUM	stellate_myofibroblast	0.624	secreted	0.10
3	4.0	DCN	stellate_myofibroblast	0.612	secreted	0.10
4	5.0	TREM2	macrophage_monocyte	0.606	surface	0.60
5	6.0	PLVAP	endothelial	0.581	surface	0.40
7	7.0	COL3A1	stellate_myofibroblast	0.553	secreted	0.20
9	8.0	MMP7	cholangiocyte	0.545	intracellular	0.10
10	9.0	SPP2	cholangiocyte	0.544	intracellular	0.10
11	10.0	MMP2	stellate_myofibroblast	0.542	secreted	0.55

Top 10 shown. The full ranked 20, with per-candidate rationale, component scores, and significance/specificity flags, is in `results/08_score/candidate_scores.tsv` and the full report.

Interpretation

Hepatic stellate / myofibroblast (PDGFRA, LUM, DCN, COL3A1, MMP2) are ECM and PDGF-axis genes consistent with activated-HSC / myofibroblast biology. **PDGFRA** is the stand-out **therapeutic** target: a cell-surface receptor tyrosine kinase already druggable with approved TKIs (imatinib / nilotinib class). The secreted matrix proteins (LUM, DCN, COL3A1, MMP2) are **diagnostic**-accessible: type III collagen turnover (COL3A1) underlies the clinically validated circulating PRO-C3 fibrosis marker, connecting this arm directly to a non-invasive readout.

Scar-associated macrophages. **TREM2** marks the CD9 / SPP1 / TREM2 pro-fibrotic macrophage program defined in this primary cohort (Ramachandran et al. 2019, *Nature*). It is dual-use: a **therapeutic** target (surface receptor, agonist antibodies in clinical development) and a **diagnostic** one (soluble TREM2 is measurable in blood), the strongest single pick for tracking the fibrotic macrophage niche.

Scar endothelium. **PLVAP** and **ACKR1** flag the disease-associated (capillarized) endothelial population, also from Ramachandran 2019; these read as mechanistic / imaging markers rather than direct drug targets.

Ductular reaction. **MMP7** and **SPP2** mark the cholangiocyte / biliary response; circulating MMP-7 is an established fibrosis / cholestasis biomarker, supporting MMP7 as a **diagnostic** candidate.

The most actionable picks are therefore **PDGFRA** (druggable target), **TREM2** (dual diagnostic / target for the macrophage niche), and the **secreted collagen / proteoglycan panel** (COL3A1, LUM, DCN) as circulating-biomarker candidates.

Calibration. Against a 37-gene panel of established human-liver-fibrosis markers, the pipeline recovers 10 as headline candidates and surfaces 25 as scored candidates, so the list's sensitivity is moderate and the gaps are cohort-driven (small-n dilution of subset markers) rather than scoring artifacts.

Limitations. The primary cohort is small (5 cirrhotic vs 5 healthy donors; 3 vs 3 for the stellate and cholangiocyte compartments), and cross-dataset reproducibility against the MASLD/MASH validation set is weak (rank correlation near zero), driven by the scRNA vs snRNA modality and disease-context mismatch rather than by the primary signal; confidence is therefore strongest for the well-powered, internally-significant stellate and macrophage hits.

Full report and source code: https://github.com/ericmalekos/sc-liver